

The SOC & Security Operations

What a SOC Does

A **Security Operations Center (SOC)** monitors, detects, investigates, and responds to cyber threats around the clock. It's the 911 dispatch + police + detective of the organization's security posture.

Core functions:

- Monitor** – watch telemetry from everywhere (endpoints, network, cloud, identity)
- Detect** – find the suspicious among the noisy
- Triage** – separate real threats from false alarms
- Investigate** – dig until there's a verdict
- Respond** – contain, eradicate, restore (with IR, module 13)
- Report & improve** – lessons, rules, signatures

The SOC Stack

Layer	Examples
SIEM	Splunk, Elastic, Sentinel, Qradar – the log brain
EDR	CrowdStrike, Defender, SentinelOne – endpoint telemetry + response
NDR/IDS/IPS	Suricata, Zeek – network detections
SOAR	playbook automation for response
Threat Intel	feeds, TI platforms – context for what's hitting you

SOC Tiers (the Career Ladder)

Tier 1 – Triage/Alert – first look at alerts, triage, escalate; the "gate"

Tier 2 – Analysts – deep investigation, containment, tuning

Tier 3 – Hunt/Experts – advanced hunting, malware deep-dive, rule writing

SOC Lead – shift management, metrics, process

Detection Engineer – builds the rules and pipelines (the craft behind it)

The Alarm Wall Problem

A SOC sees **thousands of alerts a day**. The real skill isn't answering calls – it's **reducing noise**:

Tuning / false-positive reduction

Prioritization (risk-based, not volume-based)

Good **detection engineering** (module 03 of this track)

Blue-team lens: If an alert isn't actionable, it's noise. A mature SOC measures *detection coverage* and *time-to-detect*, not alert counts.

SOC Metrics (What a SOC Is Held To)

MTTD – Mean Time to Detect

MTTR – Mean Time to Respond/Resolve

Detection coverage % (of the ATT&CK map, e.g.)

False positive rate

Escalation time

The industry's shame is A09 of OWASP and the "invisibility" of breaches for months. SOCs exist to collapse MTTD from **months to minutes**.

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